Neural Physics UAV Flight Path Optimisation in Complex Urban Environments taking into account Transient Turbulent Air Flows

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Abstract

We present a neural-physics-based framework for UAV flight path optimisation in dense urban environments with transient turbulent airflow and background advection. Unlike conventional approaches that treat wind as a disturbance, the proposed method exploits airflow structures to accelerate travel and reduce energy consumption. A wave–advection formulation is introduced to compute globally optimal paths while avoiding obstacles, terrain, and other UAVs. The approach combines ConvFEM spatial discretisation with explicit time stepping. Results demonstrate improved efficiency and scalability for city-scale deployment.

Keywords: UAV, path planning, neural physics, turbulence, optimisation, advection

1. Introduction

The rapid growth of e-commerce, on-demand logistics, and urban population density has led to increasing pressure on last-mile delivery systems. Traditional ground-based logistics networks are constrained by congestion, infrastructure limitations, and environmental concerns. As a result, unmanned aerial vehicles (UAVs), commonly referred to as drones, have emerged as a promising alternative for urban last-mile delivery and broader urban air mobility (UAM) applications.

Recent studies have highlighted the significant potential of UAV-based delivery systems to reduce congestion, lower emissions, and improve delivery efficiency, particularly for time-sensitive goods such as medical supplies, food delivery, and high-value consumer products [1],[2]. The concept of drone-assisted logistics is rapidly transitioning from experimental trials to early-stage deployment in cities worldwide, with increasing evidence suggesting that large-scale adoption is feasible under appropriate regulatory and technological frameworks [3].

In a fully realised urban drone ecosystem, it is plausible that tens or even hundreds of thousands of UAVs may be operating simultaneously within a metropolitan area. These UAVs would perform a diverse set of tasks, including:

- last-mile parcel delivery for e-commerce platforms,
- rapid delivery of medical supplies and emergency response equipment,
- infrastructure inspection and environmental monitoring,

- surveillance and public safety operations,
- dynamic data collection for smart city applications.

Such large-scale deployment introduces fundamentally new challenges in airspace management, collision avoidance, and path optimisation. The low-altitude urban airspace, often referred to as the Very Low Level (VLL) airspace, is highly constrained by buildings, terrain, and regulatory no-fly zones. Moreover, the interaction between a large number of UAVs leads to complex traffic dynamics, requiring advanced coordination mechanisms to avoid congestion and ensure safety [4, 5].

A critical challenge in this context is the computation of optimal flight paths in a dynamic, high-density environment. UAV routing must account for multiple competing objectives, including travel time, energy consumption, collision avoidance, and regulatory constraints. Furthermore, UAVs are subject to strict energy limitations due to battery capacity, making energy-efficient path planning essential for practical deployment [6].

Existing approaches to UAV path planning include graph-based methods, fast marching methods, sampling-based planners such as RRT*, and increasingly, machine learning and reinforcement learning techniques. For example, fast marching methods have been used to compute optimal paths in complex environments with obstacle avoidance [7], while reinforcement learning approaches have demonstrated the ability to learn adaptive navigation strategies in dynamic environments. However, many of these methods either neglect the influence of environmental physics or treat it as a disturbance to be mitigated.

In contrast, recent research suggests that environmental factors such as wind fields can be exploited to improve UAV performance. By aligning flight trajectories with favourable airflow, UAVs can reduce energy consumption and increase effective speed, leading to more efficient operations. This perspective represents a paradigm shift from disturbance rejection to environmental exploitation.

Another key challenge is scalability. Urban drone traffic management systems must handle extremely high traffic densities while ensuring safe separation between vehicles. Prior work has explored airspace structuring, including corridor-based routing and layered airspace designs, to reduce conflict probability and improve throughput [4]. Additionally, system-level models have been developed to analyse traffic equilibrium and congestion effects in UAV delivery networks [6].

Despite these advances, there remains a need for a unified framework that:
beginitemize
item incorporates environmental physics such as transient turbulent airflow,
item scales efficiently to city-wide deployments with large UAV fleets,
item naturally enforces obstacle avoidance and collision constraints,
item and produces time-optimal, energy-efficient flight paths.
enditemize

In this work, we address these challenges by formulating UAV flight path optimisation as a wave propagation problem augmented with environmental advection and solved using a neural physics framework. This approach

enables the computation of globally optimal paths while inherently capturing the effects of airflow, obstacles, and dynamic constraints.

The remainder of this paper introduces the mathematical formulation, numerical discretisation using neural physics ConvFEM methods, and a scalable framework for city-scale UAV optimisation.

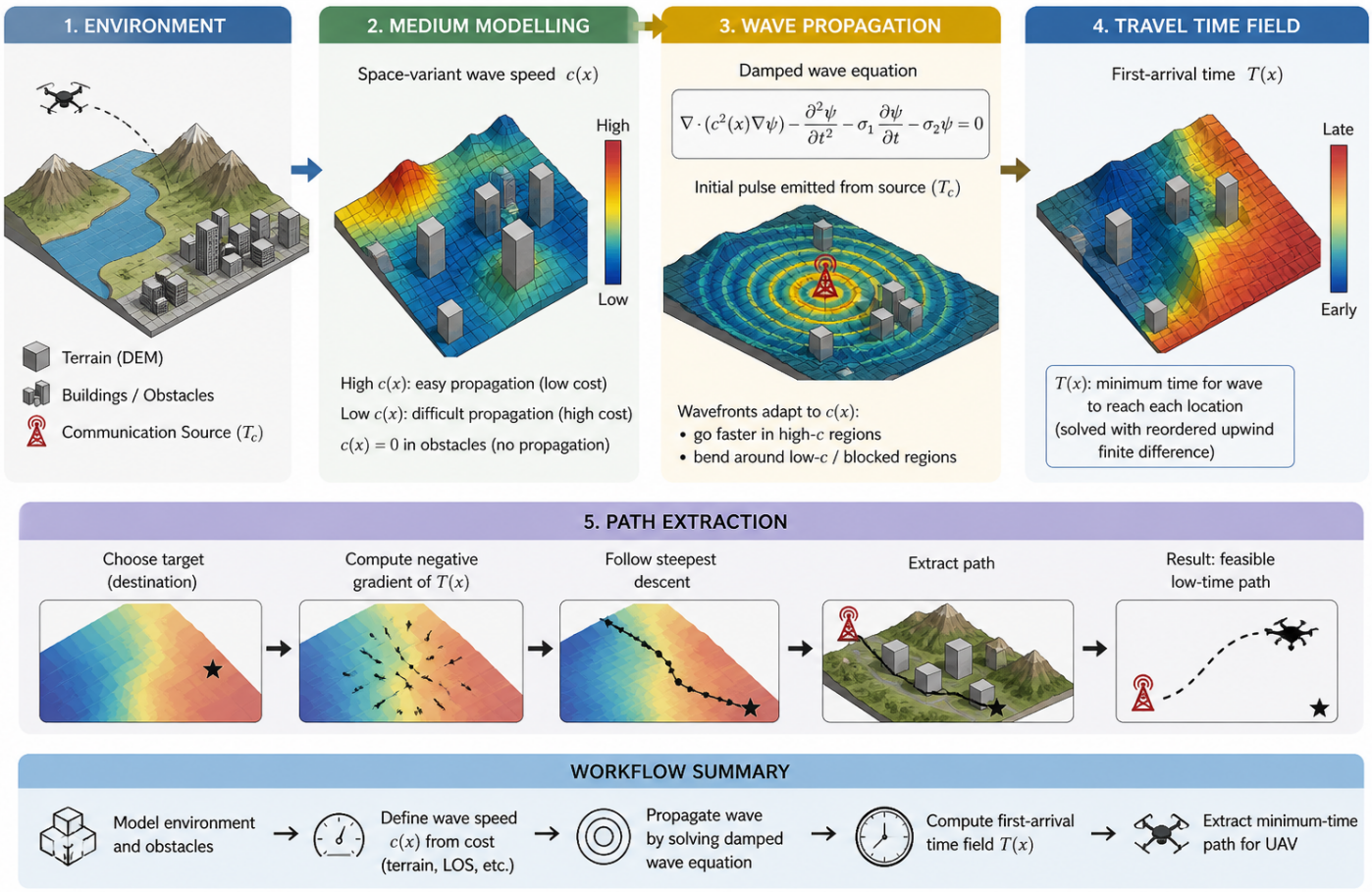


Figure 1: Schematic overview of the proposed neural physics-based UAV flight path optimisation framework. The urban environment, including terrain, buildings, and airflow, is first encoded through a spatially varying wave speed field $c(\mathbf{x})$. A wave–advection equation is then solved, allowing wavefronts to propagate while being influenced by obstacles and background wind. The resulting first-arrival-time field $T(\mathbf{x})$ represents the minimum travel time to all locations. Optimal UAV trajectories are subsequently extracted by following the negative gradient of $T(\mathbf{x})$, yielding time-efficient, obstacle-avoiding paths that naturally exploit favourable flow structures.

2. Flight Path Optimisation - UAV-Wave-Routing

Here an approach based on a UAW wave equation is developed to form the optimal path around obstacles and with a wind (air flow velocity). See Figure 1 for a summary of the overall concept.

2.1. Wave–Advection Equation Formulation

We define a scalar field $\psi(\mathbf{x}, t)$ governed by a wave equation written in terms of the material derivative:

$$\frac{D^2\psi}{Dt^2} + \sigma_1 \frac{\partial\psi}{\partial t} + \sigma_2\psi = \nabla_{xy} \cdot \left(c^2(\mathbf{x}, t)\nabla_{xy}\psi\right) + \frac{\partial}{\partial z}(c_z^2 \frac{\partial\psi}{\partial z}) + \delta(\mathbf{x} - \mathbf{x}_0)\delta(t) \tag{1}$$

in which $\nabla_{xy} = (\frac{\partial}{\partial x}, \frac{\partial}{\partial x})$ where the terms involving $_1, \sigma_2$ can be used as damping terms to form sponges near the boundary layers to avoid reflection and where the material derivative is defined as:

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \underline{q}(\mathbf{x}, t) \cdot \nabla \tag{2}$$

and thus using the background velocity $\underline{q} = (u, v, w)^T$ (containing both the air flow velocity and the vertical velocity which takes into account gravity):

$$\begin{aligned}
\frac{D^2\psi}{Dt^2} = &\frac{\partial^2\psi}{\partial t^2} + 2\left(u\frac{\partial^2\psi}{\partial x\partial t} + v\frac{\partial^2\psi}{\partial y\partial t} + w\frac{\partial^2\psi}{\partial z\partial t}\right) \\
&+ u^2\frac{\partial^2\psi}{\partial x^2} + v^2\frac{\partial^2\psi}{\partial y^2} + w^2\frac{\partial^2\psi}{\partial z^2} \\
&+ 2uv\frac{\partial^2\psi}{\partial x\partial y} + 2uw\frac{\partial^2\psi}{\partial x\partial z} + 2vw\frac{\partial^2\psi}{\partial y\partial z} \\
&+ \left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} + w\frac{\partial u}{\partial z}\right)\frac{\partial\psi}{\partial x} \\
&+ \left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} + w\frac{\partial v}{\partial z}\right)\frac{\partial\psi}{\partial y} \\
&+ \left(\frac{\partial w}{\partial t} + u\frac{\partial w}{\partial x} + v\frac{\partial w}{\partial y} + w\frac{\partial w}{\partial z}\right)\frac{\partial\psi}{\partial z}
\end{aligned} \tag{3}$$

This formulation naturally incorporates advection due to the background velocity field $\underline{q}(\mathbf{x}, t)$, biasing wave propagation along airflow directions.

The horizontal and vertical wave speeds $c(\boldsymbol{x}, t)$ and $c_z(\boldsymbol{x}, t)$ encode environmental constraints (reflecting the speed of the UAV in the horizontal and vertical):

- $c(\boldsymbol{x}, t) = c_z(\boldsymbol{x}, t) = 0$ inside buildings and ground regions
- Gaussian suppression around other UAV trajectories (setting the wave speed to zero or near zero near other UAVs):

$$c(\mathbf{x}, t) = c_0(1 - \exp\left(-\frac{||\mathbf{x} - \mathbf{x}_d(t)||^2}{\sigma^2}\right)),\ c_z(\mathbf{x}, t) = c_{z0}(1 - \exp\left(-\frac{||\mathbf{x} - \mathbf{x}_d(t)||^2}{\sigma^2}\right)). \tag{4}$$

- Modulation by turbulent flow structures

This ensures propagation avoids obstacles and other UAVs while exploiting favourable wind conditions.

2.1.1. Background Velocity and UAV Speed Definition

The background velocity is defined by

$$\boldsymbol{q} = \boldsymbol{q_{fluid}} + \boldsymbol{q_{gravity}} \tag{5}$$

in which $\boldsymbol{q_{fluid}}$ is the air-flow velocity and $\boldsymbol{q_{gravity}}$ is the velocity that embodies the force due to gravity that the UAV experiences when travelling vertically upwards and it has comonents $\boldsymbol{q_{gravity}} = (0, 0, -w_{gravity})^T$. Thus,

$c_{z0} - w_{gravity}$ is the speed in which the UAV can travel upwards and $c_{z0} + w_{gravity}$ is the speed it can travel downwards. Also c_0 represents the speed the UAV can travel in the horizontal relative to the moving fluid air.

Suppose the vertically upwards and vertically downwards UAV velocities are c_{uav-up} and $c_{uav-down}$, and the horizontal UAV speed relative to the fluid-air velocity is c_0, then the positive values of c_{z0}, $w_{gravity}$ can be defined from the solution of the two simultaneous equations:

$$c_{uav-up} = c_{z0} - w_{gravity}, \quad c_{uav-down} = c_{z0} + w_{gravity}, \tag{6}$$

and thus:

$$c_{z0} = \frac{1}{2}(c_{uav-up} + c_{uav-down}), \quad w_{gravity} = \frac{1}{2}(c_{uav-down} - c_{uav-up}). \tag{7}$$

The latter $w_{gravity} < 0$ because $c_{uav-down} < c_{uav-up}$ - that is the UAV can travel faster vertically up than vertically down for stability reasons. In addition within obsticles (buildings, topgraphy, other drones) we also suppress or relax $w_{gravity}$ to zero using for UAVs:

$$w_{gravity}(\mathbf{x}, t) = w_{gravity0}(1 - \exp\left(-\frac{||\mathbf{x} - \mathbf{x}_d(t)||^2}{\sigma^2}\right)) \tag{8}$$

and set $w_{gravity}$ to zero inb buildings and bathymetry or solid objects like vehicles, or trees.

UAV or drone speeds are typically as follows. Horizontal speed (forward) 10–20 m/s ($\approx$ 35–70 km/h) and in "sport mode": up to 25 m/s (90 km/h) for higher-end models Vertical speed Ascending (up): 3–6 m/s Descending (down): 2–5 m/s Descent is usually slower than ascent for stability and safety.

2.2. Neural Physics ConvFEM Discretisation with Advection

Expanding the material derivative in terms of the velocity components (u, v, w) gives:

$$\frac{D^2\psi}{Dt^2} = \frac{\partial^2\psi}{\partial t^2} + 2(u\partial_x + v\partial_y + w\partial_z)\left(\frac{\partial\psi}{\partial t}\right) + (u\partial_x + v\partial_y + w\partial_z)^2\psi + \underline{a} \cdot \nabla\psi \tag{9}$$

where the flow acceleration is:

$$\underline{a} = \left(\frac{\partial u}{\partial t} + \underline{q} \cdot \nabla u,\ \frac{\partial v}{\partial t} + \underline{q} \cdot \nabla v,\ \frac{\partial w}{\partial t} + \underline{q} \cdot \nabla w\right) \tag{10}$$

Spatial Discretisation (ConvFEM). The spatial operators are approximated using convolutional finite element operators. Let:

- $\boldsymbol{\psi}$: nodal representation of ψ
- $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$: discrete velocity components and thus $\boldsymbol{q} = (\boldsymbol{u}^T, \boldsymbol{v}^T, \boldsymbol{w}^T)^T$
- $\boldsymbol{A_x}, \boldsymbol{A_y}, \boldsymbol{A_z}$: convolutional gradient operators

- $\boldsymbol{A_{\nabla^2_{xy}}}$ and $\boldsymbol{A_{\nabla^2_z}}$: the convolutional Laplacian operators acting in the horizontal ($\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$) and vertical ($\frac{\partial^2}{\partial z^2}$).

First-order derivatives are computed as:

$$\partial_x \psi \;\rightarrow\; \boldsymbol{A_x \psi}, \quad \partial_y \psi \;\rightarrow\; \boldsymbol{A_y \psi}, \quad \partial_z \psi \;\rightarrow\; \boldsymbol{A_z \psi} \tag{11}$$

The advective operator becomes:

$$\underline{q} \cdot \nabla \psi \;\rightarrow\; \boldsymbol{u} \odot (\boldsymbol{A_x \psi}) + \boldsymbol{v} \odot (\boldsymbol{A_y \psi}) + \boldsymbol{w} \odot (\boldsymbol{A_z \psi}) \tag{12}$$

The second-order advective term is discretised as:

$$(\underline{q} \cdot \nabla)^2 \psi \;\rightarrow\; \boldsymbol{u} \odot \boldsymbol{A_x}(\mathbf{v} \cdot \nabla \psi) + \boldsymbol{v} \odot \boldsymbol{A_y}(\mathbf{v} \cdot \nabla \psi) + \boldsymbol{w} \odot \boldsymbol{A_z}(\mathbf{v} \cdot \nabla \psi) \tag{13}$$

Time discretisation. Using an explicit second-order time-stepping scheme, the update equation becomes:

$$\begin{aligned}
\boldsymbol{\psi}^{n+1} = {} & (2\boldsymbol{\psi}^n - \boldsymbol{\psi}^{n-1}) \\
& - \frac{1}{2}\left(\boldsymbol{A}_{\nabla^2_{xy}}(\boldsymbol{c} \odot \boldsymbol{\psi}^n) + \boldsymbol{c} \odot \boldsymbol{A}_{\nabla^2_{xy}} \boldsymbol{\psi}^n - \boldsymbol{\psi}^n \odot \boldsymbol{A}_{\nabla^2_{xy}} \boldsymbol{c}\right) \odot \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& - \frac{1}{2}\left(\boldsymbol{A}_{\nabla^2_z}(\boldsymbol{c_z} \odot \boldsymbol{\psi}^n) + \boldsymbol{c_z} \odot \boldsymbol{A}_{\nabla^2_z} \boldsymbol{\psi}^n - \boldsymbol{\psi}^n \odot \boldsymbol{A}_{\nabla^2_z} \boldsymbol{c_z}\right) \odot \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& + s^n \left(\frac{(\Delta t)^2}{\boldsymbol{M_L}}\right) \\
& - 2\Delta t \left[\boldsymbol{u}^n \odot \boldsymbol{A_x} \dot{\boldsymbol{\psi}}^n + \boldsymbol{v}^n \odot \boldsymbol{A_y} \dot{\boldsymbol{\psi}}^n + \boldsymbol{w}^n \odot \boldsymbol{A_z} \dot{\boldsymbol{\psi}}^n\right] \odot {\mathbf{M_L}}^{-1} \\
& - (\Delta t)^2 \left[\boldsymbol{u}^n \odot \boldsymbol{A_x} \mathbf{b}^n + \boldsymbol{v}^n \odot \boldsymbol{A_y} \mathbf{b}^n + \boldsymbol{w}^n \odot \boldsymbol{A_z} \mathbf{b}^n\right] \odot {\mathbf{M_L}}^{-1} \\
& - \boldsymbol{\sigma_1}(\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^n)\Delta t - \boldsymbol{\sigma_2} \boldsymbol{\psi}^{n+1} (\Delta t)^2
\end{aligned} \tag{14}$$

where $\dot{\boldsymbol{\psi}}^n$ is approximated as:

$$\dot{\boldsymbol{\psi}}^n = \frac{\boldsymbol{\psi}^{n+1} - \boldsymbol{\psi}^{n-1}}{2\Delta t}, \tag{15}$$

and

$$\mathbf{b}^n = \boldsymbol{u}^n \odot \boldsymbol{A_x} \boldsymbol{\psi}^n + \boldsymbol{v}^n \odot \boldsymbol{A_y} \boldsymbol{\psi}^n + \boldsymbol{w}^n \odot \boldsymbol{A_z} \boldsymbol{\psi}^n. \tag{16}$$

To solve Equation (14) one gathers all the coefficients associated with $\boldsymbol{\psi}^{n+1}$ to solve these equations implicitly.

2.3. First Arrival Time and Path Extraction

The scalar field $T(\mathbf{x})$ is defined as the **first arrival time** of the wavefront at each spatial location. This field is extracted from the evolving solution by recording the earliest time at which $\psi(\mathbf{x}, t)$ reaches its first peak, or local in time, maximum.

The optimal flight path $\boldsymbol{x_d}$ is obtained by following the negative gradient:

$$\frac{d\mathbf{x_d}}{ds} = \gamma_s \Delta x \frac{-\nabla T(\mathbf{x_d})}{|\nabla T(\mathbf{x_d})|}, \tag{17}$$

in which Δx is the grid spacing and γ_s is the fraction of the grid spacing used in the trajectory discretisation (e.g. $\gamma = 0.1$):

$$\mathbf{x_d}^{k+1} = \mathbf{x_d}^{k} + \gamma_s \Delta x \frac{-\nabla T(\mathbf{x_d}^{k})}{|\nabla T(\mathbf{x_d}^{k})|}, \tag{18}$$

and $\mathbf{x_d}^{k}$ represents discrete points along the UAV trajectory. This guarantees a time-optimal trajectory. The travel time is inherently encoded in $T(\mathbf{x})$, eliminating the need for separate time estimation.

2.4. Encouragement to go through certain regions in order to collect data along the journey or discouraging to go through certain regions

Encouragement to pass through certain regions can be done by increasing the values of c, c_z in those regions. This then allows the wave to travel quicker through these regions. On can still also solve for the time of flight by taking the drone along the resulting trajectory with its effective speed in the horizontal and vertical taking into account the velocity field $\underline{q}$. Thus the new field has an arrive time based on a cost T_c concept rather than time of arrival and is then used to form the path using:

$$\frac{d\mathbf{x_d}}{ds} = \gamma_s \Delta x \frac{-\nabla T_c(\mathbf{x_d})}{|\nabla T_c(\mathbf{x_d})|}, \tag{19}$$

and the descrete points along the trajectory are obtained (similar to before) from:

$$\mathbf{x_d}^{k+1} = \mathbf{x_d}^{k} + \gamma_s \Delta x \frac{-\nabla T_c(\mathbf{x_d}^{k})}{|\nabla T_c(\mathbf{x_d}^{k})|}. \tag{20}$$

One chooses the minimum path route again but now its based on cost T_c. In practice the values of c, c_z can be determined by a cost advantage of say of having a deviation of the drone to collect data.

In addition, you the drones to be near the ground level by reducing the values of c, c_z near the ground. One may have some no-go areas or we may want to discourage the drone from going through certain areas and again we can do that by reducing the values of c, c_z.

2.5. Back-propagation UAV-Wave Routing or Adjoint Routing

There may be instance were there are potentially many UAVs that can be called upon to journey to a particular location. In this case it may be more efficient to solve the wave equations from the location that needs to UAV to visit (e.g. delivery or emergency care). In this case one can still solve the wave equation but in this case one solves with the velocity $-\boldsymbol{q}$ rather than $\boldsymbol{q}$ and in addition this velocity field then propagates backwards in time. Thus we then need to solve a system with backwards from the starts of the drone journeys to form the routine:

$$\frac{d\mathbf{x_d}}{ds} = \gamma_s \Delta x \frac{\nabla T_c(\mathbf{x_d})}{|\nabla T_c(\mathbf{x_d})|}. \tag{21}$$

The only difference between this and the forward approach is the sign. In practice one may need to run the simulation for a range of ending times and the drone starting journeys. See Figure 2 to see how one can integrate the forwards and backwards UAV-wave equation modelling.

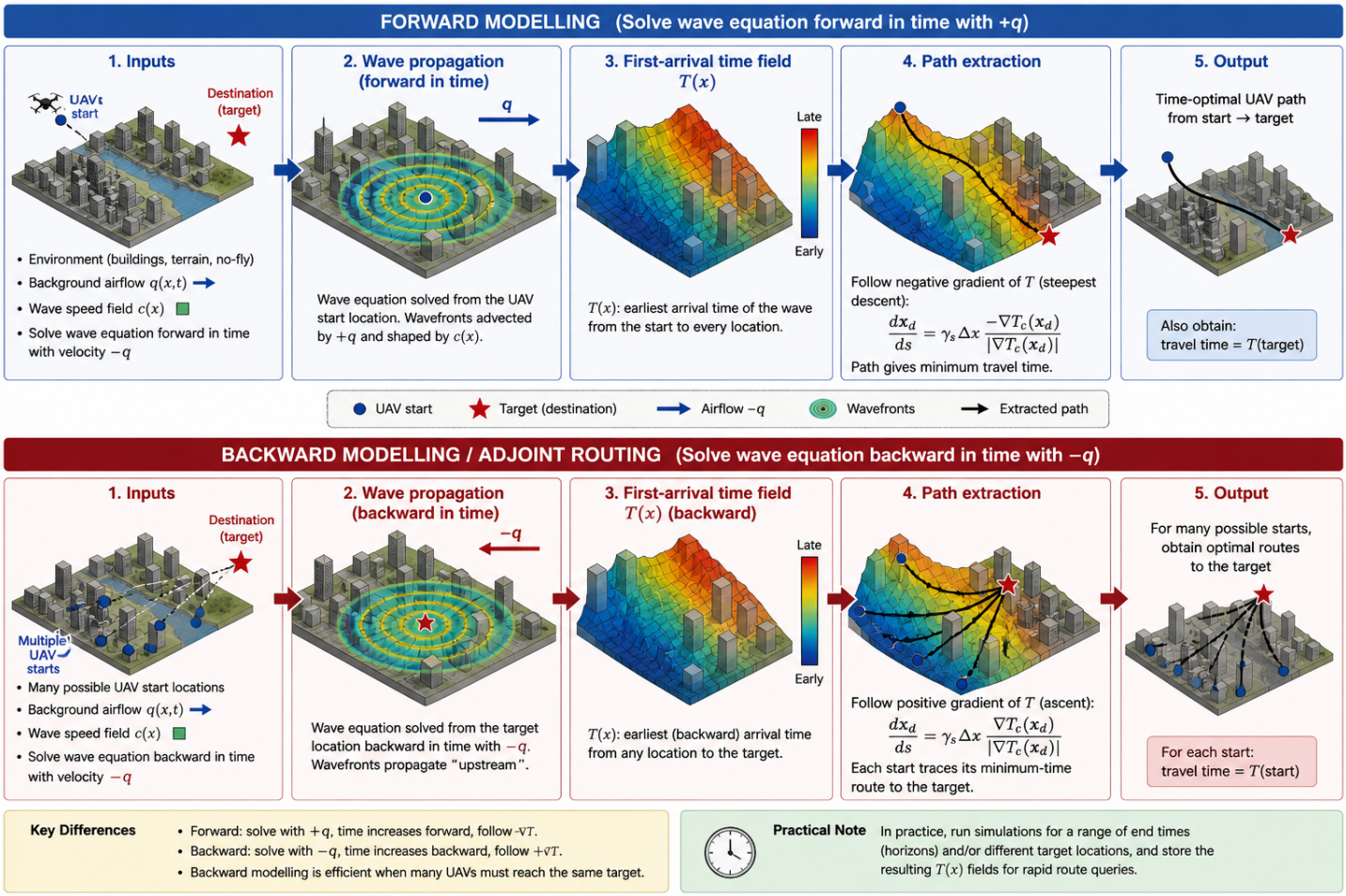


Figure 2: Schematic comparison of forward and backward (adjoint) UAV wave-based routing strategies. In the forward formulation (top), the wave equation is solved from the UAV start location and propagated forward in time under the velocity field $\boldsymbol{q}$, producing a first-arrival-time field $T(\mathbf{x})$ from which optimal trajectories are obtained by following the negative gradient $-\nabla T$. In contrast, the backward (adjoint) formulation (bottom) solves the wave equation from the target location using the reversed velocity field $-\boldsymbol{q}$, effectively propagating information backward in time. This yields a global arrival-time field that simultaneously encodes optimal routes from multiple potential UAV starting locations to the same destination, with trajectories recovered by following the positive gradient ∇T. The backward approach is particularly efficient for multi-UAV assignment problems where many agents must reach a common target.

3. UAV Dynamics and Position Evolution Along a Prescribed Path (No Wind)

Its a reasonable approximation to consider that the UAV travels at a velocity of

$$\underline{q_{uav}} = (c_0 + u, c_0 + v, c_{z0} + w)^T. \tag{22}$$

which is what we assumed with the wave equation approach. We can then track the velocity along the path of the UAV to work out its time of flight along the path. However, we can improve this by taking into account inertia as follows.

We consider the motion of a UAV of mass m travelling along a prescribed flight path in the absence of wind, i.e. $\mathbf{q} = 0$. Let the UAV position be $\mathbf{x}(t) = (x(t), y(t), z(t))^T$ and velocity $\mathbf{v} = \dot{\mathbf{x}}$.

3.1. Equation of Motion

The governing dynamics reduce to:

$$m\frac{d\mathbf{v}}{dt} = \mathbf{F}_{\text{thrust}} - \frac{1}{2}\rho C_D A|\mathbf{v}|\mathbf{v} + (0, 0, -mg)^T. \tag{23}$$

3.2. Path Parameterisation

Let the prescribed flight path be defined by a curve:

$$\mathbf{x}(s) = (x(s), y(s), z(s))^T, \tag{24}$$

where s is the arc-length parameter along the path. The unit tangent vector is:

$$\hat{\mathbf{t}}(s) = \frac{d\mathbf{x}}{ds}. \tag{25}$$

The UAV velocity can then be written as:

$$\mathbf{v}(t) = \frac{ds}{dt}\hat{\mathbf{t}}(s) = V(t)\hat{\mathbf{t}}(s), \tag{26}$$

where $V(t)$ is the scalar speed along the path.

3.3. Reduced 1D Dynamics Along the Path

Projecting the equation of motion onto the tangent direction $\hat{\mathbf{t}}$, we obtain:

$$m\frac{dV}{dt} = T - \frac{1}{2}\rho C_D A V^2 - mg\,(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z), \tag{27}$$

where $\hat{\mathbf{e}}_z = (0, 0, 1)^T$ and $(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z)$ represents the local slope of the path.

Interpretation.

- T accelerates the UAV along the path,
- drag scales with V^2 and opposes motion,
- gravity contributes depending on climb/descent:

– uphill: $(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z) > 0$ (slows UAV),

– downhill: $(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z) < 0$ (assists motion).

3.4. Power Constraint

Assuming thrust is aligned with motion:

$$T = \frac{P}{V}, \tag{28}$$

so the speed evolution becomes:

$$m\frac{dV}{dt} = \frac{P}{V} - \frac{1}{2}\rho C_D A V^2 - mg\,(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z). \tag{29}$$

3.5. Position Evolution Along the Path

The position along the path is obtained from:

$$\frac{ds}{dt} = V(t), \tag{30}$$

so that:

$$s(t) = \int_0^t V(\tau)\, d\tau. \tag{31}$$

The full 3D position is then:

$$\mathbf{x}(t) = \mathbf{x}(s(t)). \tag{32}$$

3.6. Discrete Time Integration

In numerical form, using timestep Δt:

$$V^{n+1} = V^n + \Delta t\, \frac{1}{m}\left(\frac{P}{V^n} - \frac{1}{2}\rho C_D A (V^n)^2 - mg\,(\hat{\mathbf{t}} \cdot \hat{\mathbf{e}}_z)\right), \tag{33}$$

$$s^{n+1} = s^n + \Delta t\, V^n, \tag{34}$$

$$\mathbf{x}^{n+1} = \mathbf{x}(s^{n+1}). \tag{35}$$

3.7. Connection to Wave-Based Routing

In the context of the wave-based optimisation framework, the extracted trajectory $\mathbf{x_d}(s)$ defines the path geometry, while the above equations provide a physically consistent way to:

- compute travel time along the path,
- enforce power and drag constraints,
- distinguish ascent vs descent costs.

Thus, the wave-based method determines the *optimal path*, and the present dynamics determine the *time-parametrised motion* along that path relative to the wind.

4. UAV Fleet Scheduling via Wave-Based Cost Optimisation

The wave-based framework introduced previously provides not only optimal trajectories but also a natural mechanism for global UAV fleet scheduling. By leveraging cost fields derived from backward and forward wave propagation, the multi-UAV task allocation problem can be formulated as a structured optimisation problem. See Figure 3 for a summary of how the journeys are schedualed to the drones.

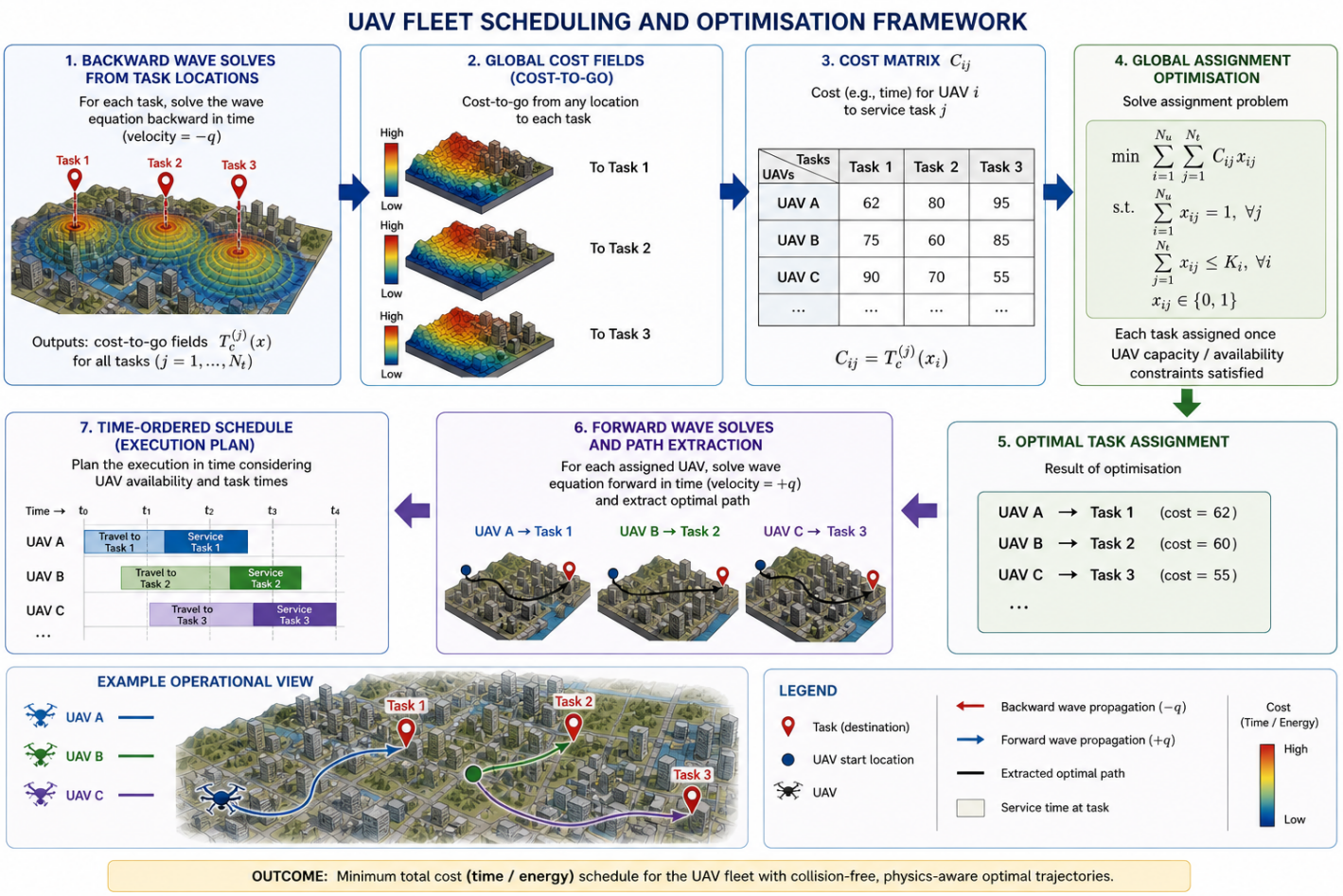


Figure 3: Schematic of the wave-based UAV scheduling framework. Backward wave propagation from task locations generates global cost-to-go fields, which are used to construct a UAV-to-task cost matrix. A global optimisation step assigns UAVs to tasks by minimising total system cost under operational constraints. Forward wave propagation is then used to compute physically consistent trajectories for each assigned UAV, resulting in a time-ordered execution schedule across the fleet.

4.1. Global Cost Field Construction

For each task location $\mathbf{x}_j^{(task)}$, a backward wave solve is performed using the reversed velocity field $-\boldsymbol{q}$. This produces a cost-to-go field:

$$T_c^{(j)}(\mathbf{x}), \tag{36}$$

which represents the travel cost from any UAV location to task j.

Given N_u UAVs located at $\{\mathbf{x}_i\}$, the pairwise cost matrix is defined as:

$$C_{ij} = T_c^{(j)}(\mathbf{x}_i), \tag{37}$$

where C_{ij} denotes the cost (time, energy, or weighted objective) for UAV i to service task j.

4.2. Assignment Optimisation

The scheduling problem is formulated as:

$$\min_{x_{ij}} \sum_{i=1}^{N_u} \sum_{j=1}^{N_t} C_{ij} x_{ij}, \tag{38}$$

subject to:

$$\sum_{i=1}^{N_u} x_{ij} = 1, \quad \forall j, \tag{39}$$

$$\sum_{j=1}^{N_t} x_{ij} \leq K_i, \quad \forall i, \tag{40}$$

$$x_{ij} \in \{0, 1\}. \tag{41}$$

Here:

- x_{ij} is the assignment variable,
- K_i is the capacity (number of tasks) UAV i can perform.

This corresponds to a classical assignment or transportation optimisation problem, but with costs derived directly from physics-based wave propagation.

4.3. Temporal Scheduling

To incorporate UAV availability and timing, we define:

$$\tilde{C}_{ij} = t_i^{(start)} + C_{ij}, \tag{42}$$

where $t_i^{(start)}$ is the earliest time UAV i can begin a task.

Additional penalties may be introduced:

$$\min \sum \tilde{C}_{ij} x_{ij} + \lambda_1 \,\text{waiting time} + \lambda_2 \,\text{congestion cost}, \tag{43}$$

allowing the framework to balance efficiency with system-level coordination.

4.4. Coupled Scheduling Interpretation

The overall framework operates as a pipeline:

1. Backward wave solves generate global cost fields,

2. Costs are assembled into a matrix for all UAV-task pairs,
3. A global optimisation assigns UAVs to tasks,
4. Forward wave solves generate executable trajectories,
5. A time-ordered schedule is produced for deployment.

4.5. Advantages

- **Scalable:** avoids pairwise path computation,
- **Globally optimal:** minimises total system cost,
- **Physics-consistent:** all costs derived from wave dynamics,
- **Flexible:** supports time, energy, and multi-objective optimisation.

This unified formulation enables real-time coordination of large UAV fleets operating in complex and dynamic urban environments.

5. Mean-Field Neural Network Formulation for UAV Scheduling Optimisation

We now formulate the UAV scheduling problem as a continuous optimisation problem using a mean-field neural network approach. This enables scalable assignment of a large number of UAVs to tasks while minimising total operational cost, including travel time, energy expenditure, and congestion.

5.1. Probabilistic assignment variables

Let:

- $i \in \{1, \ldots, N\}$ index UAVs,
- $\mu \in \{1, \ldots, M\}$ index tasks (deliveries, sensing missions, etc.).

We introduce continuous variables:

$$x_i^\mu \in [0, 1], \tag{44}$$

representing the probability that UAV i is assigned to task μ.

These satisfy the normalisation constraint:

$$\sum_{\mu=1}^{M} x_i^\mu = 1, \quad \forall i, \tag{45}$$

ensuring that each UAV is assigned to exactly one task in the discrete limit.

5.2. Cost functional

The scheduling problem is formulated as minimisation of a functional:

$$F = F_{\text{cost}} + F_{\text{balance}}. \tag{46}$$

(i) Transport and interaction cost. Let C_{ij} denote the cost of assigning UAVs i and j to interacting tasks (e.g. proximity conflicts, airspace congestion), derived from the wave-based travel-time fields $T(\mathbf{x})$ or $T_c(\mathbf{x})$.

Define:

$$F_{\text{cost}} = \frac{1}{2} \sum_{\mu=1}^{M} \sum_{i,j=1}^{N} x_i^\mu C_{ij} x_j^\mu. \tag{47}$$

This penalises assignments that cluster UAVs in regions of high congestion or overlapping trajectories.

(ii) Load balancing term. To distribute tasks evenly across UAVs, we introduce:

$$F_{\text{balance}} = \frac{\lambda}{2} \sum_{\mu=1}^{M} \left(\sum_{i=1}^{N} x_i^\mu - \frac{N}{M} \right)^2, \tag{48}$$

where λ is a weighting parameter.

This term enforces equitable allocation of UAV resources across tasks.

(iii) Combined functional. The total functional becomes:

$$F = \frac{1}{2} \sum_{\mu} x^{\mu T} C x^\mu + \frac{\lambda}{2} \sum_{\mu} \left(\mathbf{1}^T x^\mu - \frac{N}{M} \right)^2. \tag{49}$$

5.3. Mean-field equations

Define the mean-field (local cost) for each assignment:

$$f_i^\mu = \frac{\partial F}{\partial x_i^\mu}. \tag{50}$$

This gives:

$$f_i^\mu = \sum_{j=1}^{N} C_{ij} x_j^\mu + \lambda \left(\sum_{j=1}^{N} x_j^\mu - \frac{N}{M} \right). \tag{51}$$

The probabilistic assignment is then updated using a Boltzmann distribution:

$$x_i^\mu = \frac{\exp\left(-f_i^\mu / T\right)}{\sum_{\nu=1}^{M} \exp\left(-f_i^\nu / T\right)}, \tag{52}$$

where T is a temperature parameter.

5.4. Annealing and convergence

The system is solved iteratively while gradually reducing T:

- High T: diffuse assignments ($x_i^\mu \approx 1/M$),
- Low T: discrete assignments ($x_i^\mu \to 0$ or 1).

This simulated annealing process avoids poor local minima and yields near-optimal scheduling.

5.5. Critical temperature analysis

The critical temperature T_c determines when the uniform solution becomes unstable.

Assume an initial uniform state:

$$x_i^\mu = \frac{1}{M}. \tag{53}$$

Introduce a perturbation δx_i^μ and linearise the update. Growth of perturbations occurs when:

$$\frac{1}{T}\sum_{j=1}^{N} C_{ij} > 1. \tag{54}$$

Thus, the local critical temperature is:

$$T_c^{(i)} = \frac{1}{M}\sum_{j=1}^{N} C_{ij}. \tag{55}$$

A global critical temperature can be estimated as:

$$T_c = \frac{1}{NM}\sum_{i=1}^{N}\sum_{j=1}^{N} C_{ij}. \tag{56}$$

Below T_c, the system transitions from a uniform assignment to structured scheduling patterns.

5.6. Integration with UAV wave-based costs

The cost matrix C_{ij} can be constructed using:

- differences in arrival times $T(\mathbf{x})$,
- overlap of optimal trajectories,
- energy consumption derived from the velocity field $\boldsymbol{q}$,
- congestion penalties in high-density regions.

This ensures full consistency between:

- low-level path optimisation (wave propagation),
- high-level scheduling (neural mean-field optimisation).

5.7. Summary

The proposed mean-field neural formulation provides:

- scalable assignment for large UAV fleets,
- continuous optimisation with discrete convergence,
- natural integration with physics-based cost fields,
- efficient GPU-parallel implementation.

This framework enables real-time scheduling of UAVs in complex urban environments while accounting for both physical constraints and system-level optimisation objectives.

6. City-Scale Parallelisation

The urban domain is discretised into a 3D grid and partitioned into red–black subdomains.

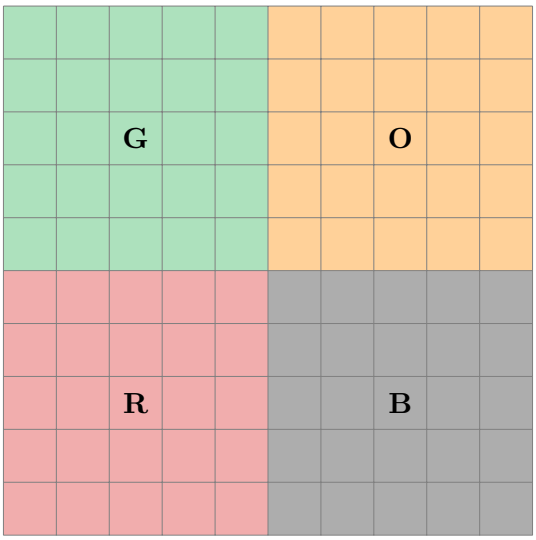


Figure 4: Uniform grid partitioned into four coloured subdomains (R, B, G, O) for parallelised UAV path optimisation using domain decomposition.

Each subdomain can be solved independently, enabling large-scale parallelisation.

Efficient deployment of the proposed wave-based UAV optimisation framework at city scale requires the ability to solve large three-dimensional problems in real time. To achieve this, we adopt a multi-colour domain decomposition strategy combined with the local stencil structure of the ConvFEM discretisation.

6.1. Domain decomposition strategy

The urban computational domain Ω is discretised into a structured or semi-structured grid and partitioned into a set of non-overlapping subdomains:

$$\Omega = \bigcup_{k=1}^{N_s} \Omega_k, \tag{57}$$

where each subdomain Ω_k is assigned a colour from a finite set (e.g. red, black, green, orange). The colouring is constructed such that no two adjacent subdomains share the same colour.

This multi-colour partitioning enables parallel updates without data conflicts, as neighbouring regions are guaranteed not to be updated simultaneously. As a result, all subdomains of the same colour can be processed concurrently.

6.2. Parallel ConvFEM wave propagation

The ConvFEM formulation expresses spatial operators as local convolutional stencils. This locality is key to parallelisation, as updates at each grid point depend only on nearby values. For a given time step, the update proceeds colour-by-colour:

1. Select a colour group $\mathcal{C}_m$,
2. Update all subdomains $\Omega_k \in \mathcal{C}_m$ in parallel,
3. Synchronise boundary information,
4. Proceed to the next colour.

This approach is well suited to GPU and distributed architectures, where each subdomain can be assigned to a processing unit. The result is near-linear scaling with the number of computational resources.

6.3. Integration with flow physics

A critical aspect of the framework is that the domain decomposition does not decouple the underlying physics. The velocity field $\boldsymbol{q}(\mathbf{x}, t)$, which represents the airflow and gravitational effects, is defined continuously across the domain and is incorporated directly into the wave equation via the material derivative.

At subdomain interfaces, consistency is maintained through boundary exchange of:

- wave field values ψ,
- velocity components (u, v, w),
- auxiliary quantities such as gradients and fluxes.

This ensures that advective transport, wave propagation, and flow-driven effects remain physically consistent across subdomain boundaries.

6.4. Handling heterogeneous urban environments

Urban environments exhibit strong spatial heterogeneity due to buildings, terrain, and local flow structures. These effects are encoded in:

- spatially varying wave speeds $c(\mathbf{x}, t), c_z(\mathbf{x}, t)$,
- obstacle masks (zero-speed regions),

- turbulent and transient velocity fields.

The domain decomposition naturally accommodates this heterogeneity, as each subdomain can operate with locally defined coefficients and constraints. This enables efficient modelling of complex features such as urban canyons, wake regions, and vertical flow structures.

6.5. Parallel multi-source simulations

An additional advantage of the wave-based formulation is the ability to run multiple simulations simultaneously across the domain. Since wave solutions decay away from their source regions (especially with damping terms), different routing problems can be solved in parallel with minimal interference.

This allows:

- simultaneous path planning for multiple UAVs,
- concurrent backward solves for multiple tasks,
- efficient utilisation of computational resources across the entire city domain.

6.6. Scalability and performance

Let N denote the total number of grid points and P the number of processors. Under ideal conditions, the computational cost per time step scales as:

$$\mathcal{O}\left(\frac{N}{P}\right), \tag{58}$$

due to the locality of stencil operations.

The multi-colour decomposition introduces minimal overhead while enabling high levels of concurrency. Combined with GPU acceleration, this allows real-time or near-real-time simulation for city-scale domains involving millions of degrees of freedom.

6.7. Summary

The multi-colour domain decomposition framework provides a scalable and efficient mechanism for solving the wave-based UAV optimisation problem at city scale. By leveraging the locality of ConvFEM operators and maintaining consistency of the underlying flow physics, the method enables large-scale, high-fidelity simulations that are suitable for real-time urban air mobility applications.

7. Discussion

The proposed neural physics-based UAV flight path optimisation framework introduces a unified formulation in which environmental physics, numerical discretisation, and data-driven correction mechanisms are tightly integrated. This represents a shift away from traditional graph-based or sampling-based planners toward a continuous physics-driven representation of optimal navigation in complex urban airspace.

7.1. Exploitation of atmospheric dynamics

A key advantage of the framework is the explicit exploitation of wind and advective flow structures. Rather than treating atmospheric motion as a disturbance to be mitigated, the formulation embeds the background velocity field (u, v, w) directly into the governing dynamics via the material derivative. This allows UAV trajectories to naturally align with favourable flow structures, effectively enabling vehicles to exploit or vertically rising convection plumes or urban wind corridors formed by building geometry. In dense urban environments, such flow channeling effects are particularly pronounced, producing locally accelerated regions between high-rise structures. By incorporating these effects into the optimisation process, UAVs can reduce both flight time and energy consumption, which is critical for battery-limited aerial systems.

7.2. Obstacle avoidance and collision modelling

Obstacle avoidance is enforced through a spatially varying wave speed fields $c(\mathbf{x}, t)$ and $c_z(\mathbf{x}, t)$, where $c = c_z = 0$ inside buildings and terrain regions as well as $w_{gravity} = 0$ in these volumes. This ensures that wave propagation is strictly confined to feasible flight corridors without requiring explicit geometric collision checks. In addition, in a similar way inter-UAV collision avoidance is handled through dynamic modulation of $c(\mathbf{x}, t)$ using Gaussian suppression fields centred on other UAV trajectories. This transforms multi-agent collision avoidance into a continuous field modification problem, avoiding combinatorial complexity associated with pairwise trajectory optimisation. As a result, the framework remains scalable even in extremely dense airspace scenarios.

7.3. First arrival time representation

A central feature of the formulation is the use of the first arrival time field $T(\mathbf{x})$, which encodes the globally optimal travel time from a source location. Unlike traditional trajectory optimisation methods that compute paths explicitly, the solution is embedded implicitly in the evolution of a scalar wave field. Optimal UAV trajectories are recovered via gradient descent on the arrival time field:

$$\frac{d\mathbf{x_d}}{ds} = -\nabla T(\mathbf{x}),$$

or time to costs:

$$\frac{d\mathbf{x_d}}{ds} = -\nabla T_c(\mathbf{x}),$$

ensuring that paths are time-optimal under the governing physics model. Importantly, temporal evolution is inherently encoded in $T(\mathbf{x})$, eliminating the need for explicit time parameterisation during path extraction if one optimises for a path based on minimising time to arrival.

7.4. UAV Scheduler

The UAV scheduler is responsible for assigning available UAVs to tasks and determining the execution order based on the wave-derived cost information. It operates on the cost matrix C_{ij}, where each entry represents the travel cost from UAV i to task j, computed from the backward wave-based cost fields.

The scheduler solves a constrained optimisation problem to minimise the total system cost while ensuring that each task is assigned and UAV operational limits are respected. This includes constraints on UAV availability, maximum number of tasks per UAV, and temporal considerations such as start times or deadlines.

In practice, the scheduler can be implemented using efficient algorithms for assignment problems (e.g. Hungarian method or linear programming relaxations), enabling real-time performance even for large fleets. The output of the scheduler is a mapping between UAVs and tasks, along with an ordered sequence of task execution where required.

Once scheduling is complete, each assigned UAV computes its trajectory using the forward wave-based optimisation framework. This separation between scheduling (global decision-making) and trajectory generation (local physics-based optimisation) ensures both scalability and physical consistency.

Overall, the UAV scheduler acts as the central coordination layer, linking global cost information with executable flight plans, and enabling efficient operation of large-scale UAV systems in complex urban environments.

7.5. Scalability and domain decomposition

Scalability is achieved through ConvFEM discretisation and domain decomposition strategies. The convolutional finite element representation transforms differential operators into local stencil computations, which are highly efficient on GPU architectures.

A multi-colour decomposition of the urban domain allows independent subregions to be solved in parallel. This is particularly important for city-scale deployments involving up to 10^5 UAVs, where real-time performance is required. The reason why this works is that the waves decay sufficiently far from the region of interest enabling one to perform the simulations across the entire city domain at the same time without these simulations interfearing with one another.

7.6. Computational complexity

Let N denote the number of grid points in the discretised domain. The ConvFEM formulation results in a computational complexity of approximately:

$$\mathcal{O}(N)$$

per time step, due to local convolutional stencil operations.

With T time steps, the total complexity becomes:

$$\mathcal{O}(NT),$$

which is highly scalable under parallel GPU execution.

In contrast:

- Dijkstra-based methods scale as $\mathcal{O}(N \log N)$,

- RRT* methods exhibit stochastic and non-deterministic convergence costs,
- Reinforcement learning approaches require costly offline training phases.

Thus, the proposed method is particularly advantageous for real-time large-scale deployment.

7.7. Comparison with existing methods

Method	Optimality	Scalability	Physics awareness
A* / Dijkstra	Exact (grid)	Moderate	None
RRT*	Asymptotic	Low–Moderate	None
Fast Marching	Approximate optimal	Moderate	Partial
Reinforcement Learning	Approximate	High (after training)	Learned only
This work	Physics-optimal	High (GPU scalable)	Full physics coupling

Table 1: Comparison of UAV path planning approaches.

7.8. Real-world deployment considerations

For practical deployment, integration with Urban Air Traffic Management (UTM) systems is required. This includes compliance with regulatory frameworks such as those proposed by the FAA (United States) and EASA (Europe), which impose constraints on altitude, geofencing, and no-fly zones.

The proposed framework is compatible with such systems by encoding regulatory constraints directly into the wave speed field $c(\mathbf{x}, t)$ and boundary conditions. This allows regulatory enforcement to be embedded natively into the optimisation process rather than applied as a post-processing step.

Additionally, real-world implementation will require coupling with real-time meteorological data and possibly reduced-order surrogate models for computational efficiency in live systems.

7.9. Limitations

Despite its advantages, several limitations remain. The accuracy of the method is strongly dependent on the fidelity of the wind field (u, v, w), which in practice may be uncertain or sparsely observed. Errors in flow estimation can propagate into suboptimal path generation. Furthermore, while neural corrections improve robustness, they introduce additional model complexity. Finally, extreme urban density scenarios may require fine spatial resolution as well as correspondily fine temporal resolution.

7.10. Summary

Overall, the framework demonstrates how wave-based physics, neural operator learning, and finite element discretisation can be unified into a scalable paradigm for UAV navigation. It provides a physically grounded yet computationally efficient approach for managing large-scale autonomous aerial systems in complex urban environments.

A schematic of the first-arrival-time field and resulting trajectory extraction process is illustrated in Fig.5.

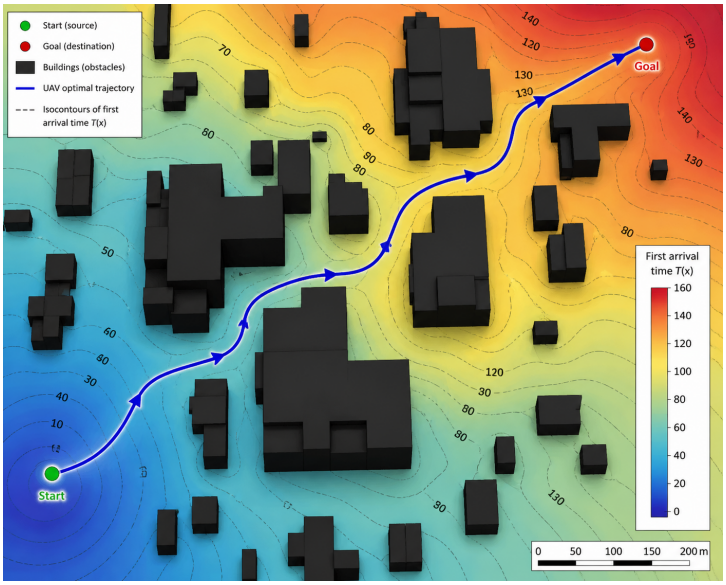


Figure 5: Obstacle-aware UAV trajectory extracted from first-arrival-time field. Buildings enforce zero wave speed $c(\mathbf{x}) = 0$, producing a physically consistent path that circumvents all obstacles.

8. Conclusion

In this work, we have presented a novel neural physics-based framework for UAV flight path optimisation in complex urban environments characterised by transient turbulent airflow and high obstacle density. By reformulating the routing problem as a wave–advection system governed by the material derivative, the proposed approach fundamentally integrates environmental physics into the optimisation process rather than treating it as an external disturbance.

A key contribution of this framework is the use of the first-arrival-time field $T(\mathbf{x})$, which implicitly encodes globally optimal trajectories. This eliminates the need for explicit path search algorithms and enables efficient extraction of time-optimal routes via gradient descent. The incorporation of airflow through the background velocity field $\mathbf{q}(\mathbf{x}, t)$ allows UAVs to exploit favourable wind structures, leading to improvements in both travel time and energy efficiency.

The use of ConvFEM discretisation provides a scalable and computationally efficient numerical backbone, particularly well-suited for GPU acceleration and large-scale deployment. Furthermore, the integration of neural operator corrections enhances the model's ability to capture complex turbulent interactions and mitigate numerical artefacts, bridging the gap between classical physics-based modelling and data-driven approaches.

The framework naturally enforces obstacle avoidance and inter-UAV collision constraints through spatial modulation of wave speeds, avoiding the need for combinatorial collision checking. In addition, the hierarchical routing strategy enables scalable coordination across large UAV fleets, supporting both global optimisation and local adaptability in dynamic environments.

Despite these advances, several challenges remain. The performance of the method is dependent on the accuracy and availability of high-resolution wind field data, and further work is required to robustly handle uncertainty in environmental inputs. Additionally, real-time deployment will require efficient data assimilation, reduced-order

modelling, and integration with existing urban air traffic management systems.

Future research directions include:

- coupling with real-time meteorological data streams and urban flow simulations,
- development of uncertainty-aware and probabilistic formulations,
- integration with reinforcement learning for adaptive policy refinement,
- and large-scale validation in realistic city environments with dense UAV traffic.

Overall, this work demonstrates that neural physics-based wave formulations provide a powerful and scalable paradigm for UAV navigation. By unifying physical modelling, numerical methods, and machine learning, the proposed approach offers a promising pathway toward efficient, safe, and intelligent urban air mobility systems at city scale.

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Algorithm 1 Neural Physics-Based UAV Flight Path Optimisation in Urban Environments

Require: Urban domain Ω, building mask Ω_b, wind field (u, v, w), wave speed $c(\mathbf{x}, t)$, source location $\mathbf{x}_0$, destination $\mathbf{x}_d$

Ensure: Optimal UAV trajectory Γ and arrival time field $T(\mathbf{x})$

Initialise $\boldsymbol{\psi}^0 = 0$, $\boldsymbol{\psi}^{-1} = 0$

Set initial wave source: $s(\mathbf{x}, t) = \delta(\mathbf{x} - \mathbf{x}_0)\delta(t)$

Construct ConvFEM operators $\boldsymbol{A}_\nabla$ and $\boldsymbol{A}_{\nabla^2}$

Apply constraints:

- $c(\mathbf{x}, t) = 0$ for $\mathbf{x} \in \Omega_b$ (buildings/ground)
- $c(\mathbf{x}, t) \to 0$ near other UAV trajectories

for $n = 1$ to N_t **do**

Compute velocity-advection terms:

$$\mathbf{b}^n = \boldsymbol{u} \odot \boldsymbol{A_x}\boldsymbol{\psi}^n + \boldsymbol{v} \odot \boldsymbol{A_y}\boldsymbol{\psi}^n + \boldsymbol{w} \odot \boldsymbol{A_z}\boldsymbol{\psi}^n$$

Update wave field using ConvFEM:

$\boldsymbol{\psi}^{n+1} \leftarrow \mathcal{F}_{\text{ConvFEM}}(\boldsymbol{\psi}^n, \boldsymbol{\psi}^{n-1}, c, u, v, w, s^n)$

Apply neural correction operator:

$$\boldsymbol{\psi}^{n+1} \leftarrow \boldsymbol{\psi}^{n+1} + \mathcal{N}_\theta(\boldsymbol{\psi}^n, \nabla\boldsymbol{\psi}^n)$$

Enforce boundary constraints:

$$\boldsymbol{\psi}^{n+1}(\mathbf{x} \in \Omega_b) = 0$$

Compute first arrival time:

$$T(\mathbf{x}) = \min(T(\mathbf{x}), t_n \mid \psi(\mathbf{x}, t_n) > \tau)$$

Extract optimal trajectory:

Initialise $\mathbf{x} \leftarrow \mathbf{x}_d$

for $k = 1$ to N_{path} **do**

Compute descent direction:

$$\mathbf{d} = -\nabla T(\mathbf{x})$$

Update position:

$$\mathbf{x} \leftarrow \mathbf{x} + \epsilon\mathbf{d}$$

Append $\mathbf{x}$ to trajectory Γ

return Γ, $T(\mathbf{x})$